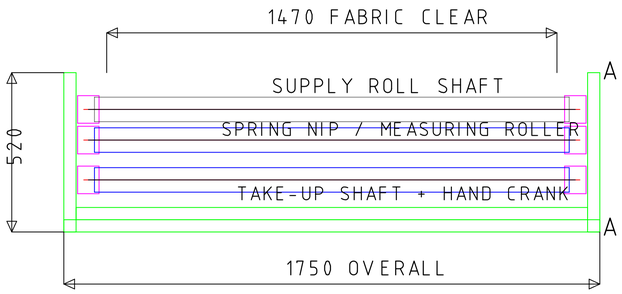
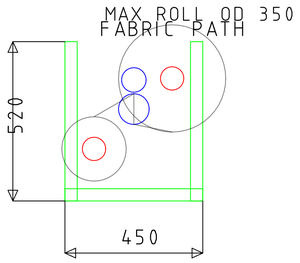
FABRIC ROLL LENGTH METER - GENERAL ARRANGEMENT

PRELIMINARY FABRICATION DRAWING | UNITS: mm | SCALE: 1:10 ON A1

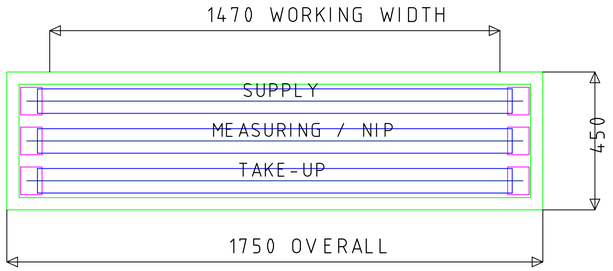
FRONT ELEVATION



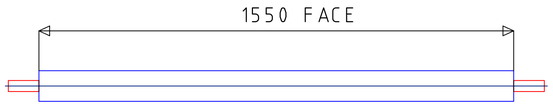
SIDE SECTION A-A



TOP VIEW

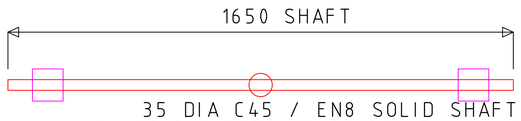


MEASURING ROLLER DETAIL



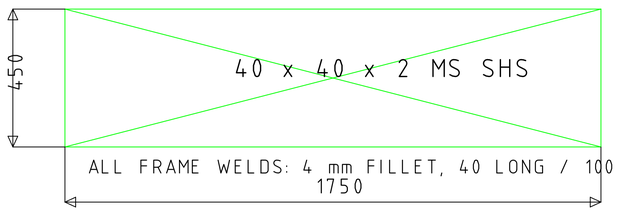
100 OD RUBBER-LAGGED ALUMINIUM ROLLER
35 DIA SHAFT ENDS | ENCODER DIRECT-COUPLED
ROLLER RUNOUT <= 0.20 TIR

SUPPLY / TAKE-UP SHAFT DETAIL



35 DIA C45 / EN8 SOLID SHAFT
ADJUSTABLE CORE CONES FOR 76 mm NOMINAL CORE

FRAME / WELDMENT



ALL FRAME WELDS: 4 mm FILLET, 40 LONG / 100 PITCH UNLESS NOTED

BILL OF MATERIALS

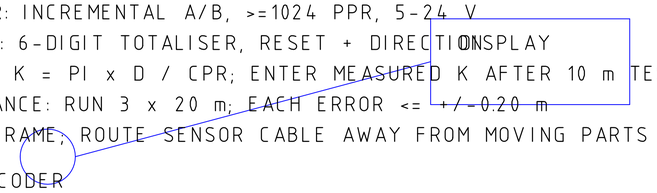
ITEM	DESCRIPTION	QTY	MATERIAL	REMARK
1	BASE FRAME WELDMENT	1	MS SHS	40x40x2
2	SUPPLY SHAFT	1	EN8	35 DIA x 1650
3	TAKE-UP SHAFT	1	EN8	35 DIA x 1650
4	MEASURING ROLLER	1	AL+RUBBER	100 OD x 1550
5	NIP PRESSURE ROLLER	1	AL+RUBBER	80 OD x 1550
6	PILLOW BLOCK UCP207	6	BOUGHT	35 mm BORE
7	ROTARY ENCODER	1	BOUGHT	>=1024 PPR
8	SPRING ADJUSTER	2	MS/SPRING	25-80 N EACH
9	HAND CRANK + CLUTCH	1	BOUGHT	FOLDING HANDLE
10	GUARDS / FEET / FASTENERS	1 SET	MIXED	AS REQUIRED

DESIGN / FABRICATION NOTES

1. DESIGN BASIS: 1470 FABRIC WIDTH, 40 kg MAX ROLL, 350 MAX OD, 76 NOMINAL CORE.
2. SUPPORT SHAFTS IN UCP207 BEARINGS; ALIGN BEARING CENTRES WITHIN 0.5 mm.
3. MEASURING ROLLER CONTACTS FABRIC THROUGH SPRING NIP; PREVENT FABRIC SLIP.
4. CALIBRATE: metres = encoder_counts x roller_circumference / counts_per_rev.
5. TARGET <1% REQUIRES ROLLER OD CALIBRATION AND A MEASURED SLIP TEST.
6. GUARD ALL PINCH POINTS WITH REMOVABLE 1.5 mm SHEET-METAL COVERS.
7. DEBURR; PRIME AND POWDER COAT FRAME AFTER TRIAL ASSEMBLY.
8. GENERAL TOLERANCE +/-1 mm; MACHINED SHAFT FEATURES +/-0.05 mm.
9. VERIFY ROLL OD, CORE, WEB TENSION AND FLOOR SPACE BEFORE RELEASE.
10. PRELIMINARY REV A: FABRICATOR / ENGINEER MUST VALIDATE LOADS BEFORE BUILD.

COUNTER / CALIBRATION

ENCODER: INCREMENTAL A/B, >=1024 PPR, 5-24 V
DISPLAY: 6-DIGIT TOTALISER, RESET + DIRECTION
NOMINAL K = PI x D / CPR; ENTER MEASURED K AFTER 10 m TEST
ACCEPTANCE: RUN 3 x 20 m; EACH ERROR <= +/-0.20 m
EARTH FRAME; ROUTE SENSOR CABLE AWAY FROM MOVING PARTS



FABRIC ROLL LENGTH METER GENERAL ARRANGEMENT & FABRICATION	REV A UNITS mm	A1 2026-10-16
DRAWING: FRM-1470-GA-001 STATUS: PRELIMINARY / NOT FOR PRODUCTION	SCALE 1:10	OPEN CAD STUDIO